

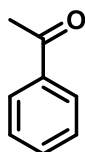
Chem 0345

Activity 9: EAS

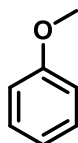
Extra Practice Problems

- 1) Draw resonance structures of the following compounds. Specify if the substituent is an electron donating group (EDG) or electron withdrawing group (EWG) with respect to the benzene ring.

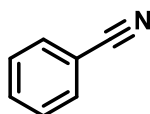
A.



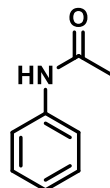
B.



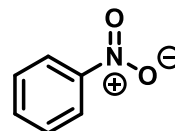
C.



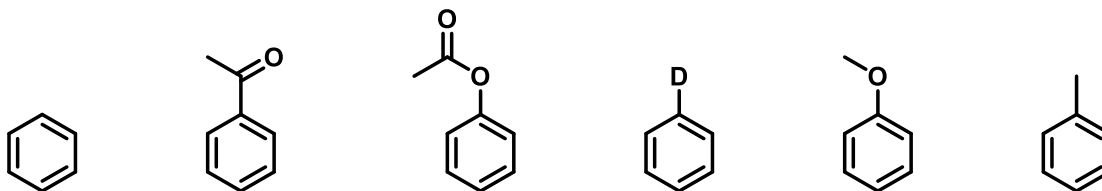
D.



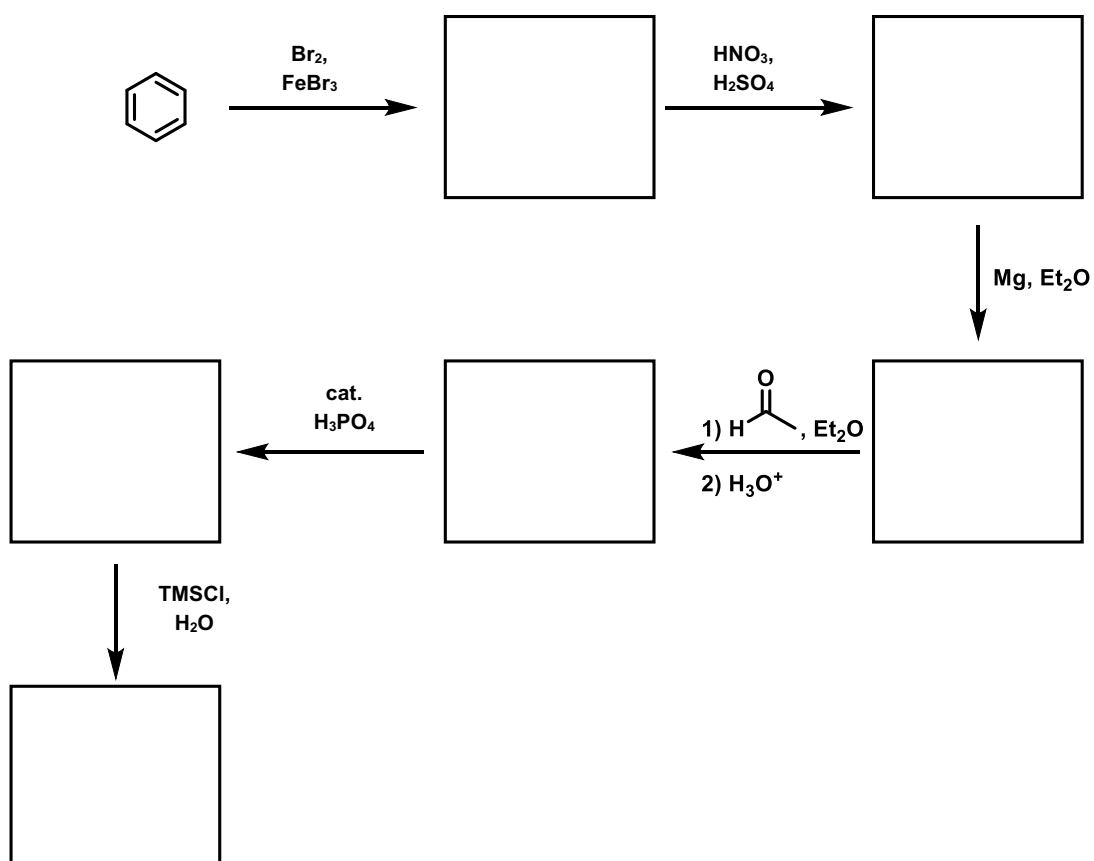
E.



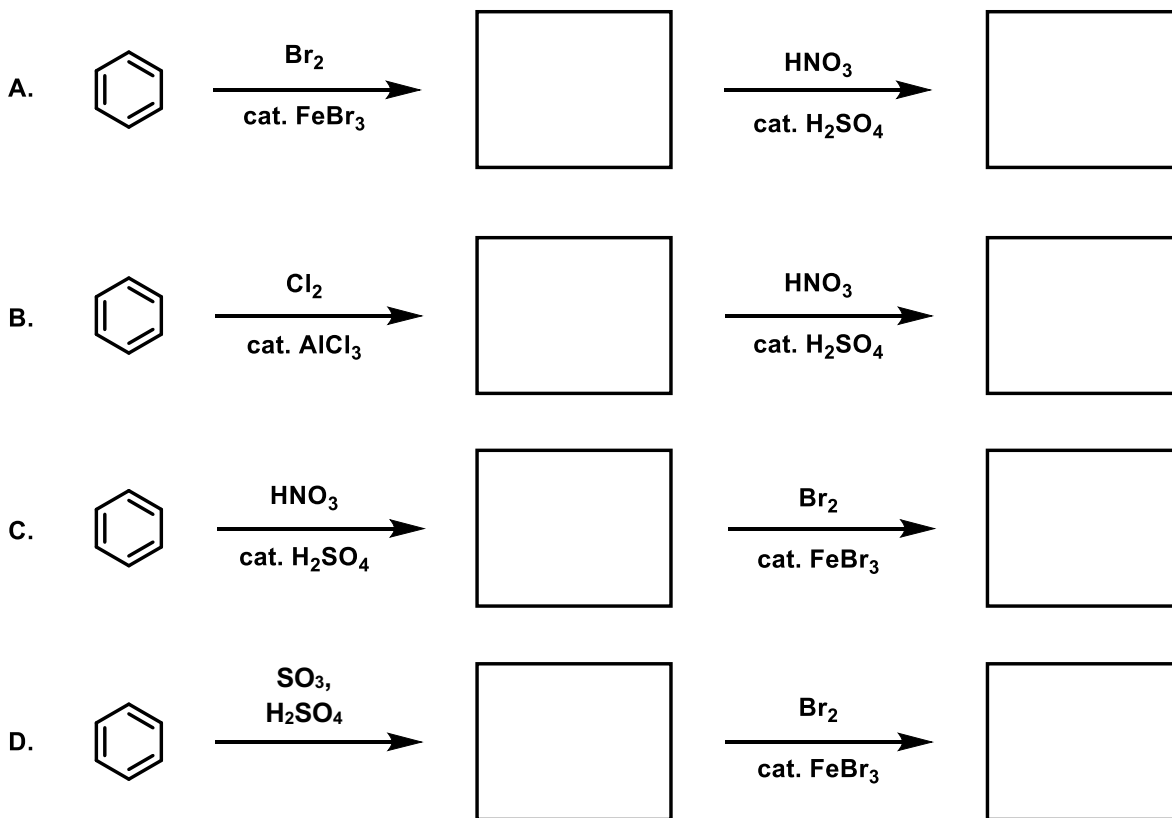
- 2) Rank the following in following molecules in terms of reacting with Cl_2 and AlCl_3 . 1 is the fastest, 5 is the slowest. (Two of these molecules have the same rate of reactivity.)



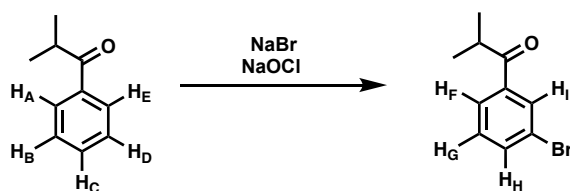
- 3) Fill in the boxes with the major deuterium-containing compound.



4) Predict the major product of the following reactions.



5) The following EAS reaction is performed and a ^1H NMR spectrum of the reaction mixture is obtained in CD_3OD :



A. Predict the splitting pattern of each proton in the starting material and product. Assume all 3-bond and 4-bond aromatic coupling interactions are observed. Assume all $^3J_{\text{Aromatic}} > ^4J_{\text{Aromatic}}$. You can also assume that all 3J are equivalent to each other and all 4J are equivalent to each other.

$\text{H}_\text{A} =$

$\text{H}_\text{F} =$

$\text{H}_\text{B} =$

$\text{H}_\text{G} =$

$\text{H}_\text{C} =$

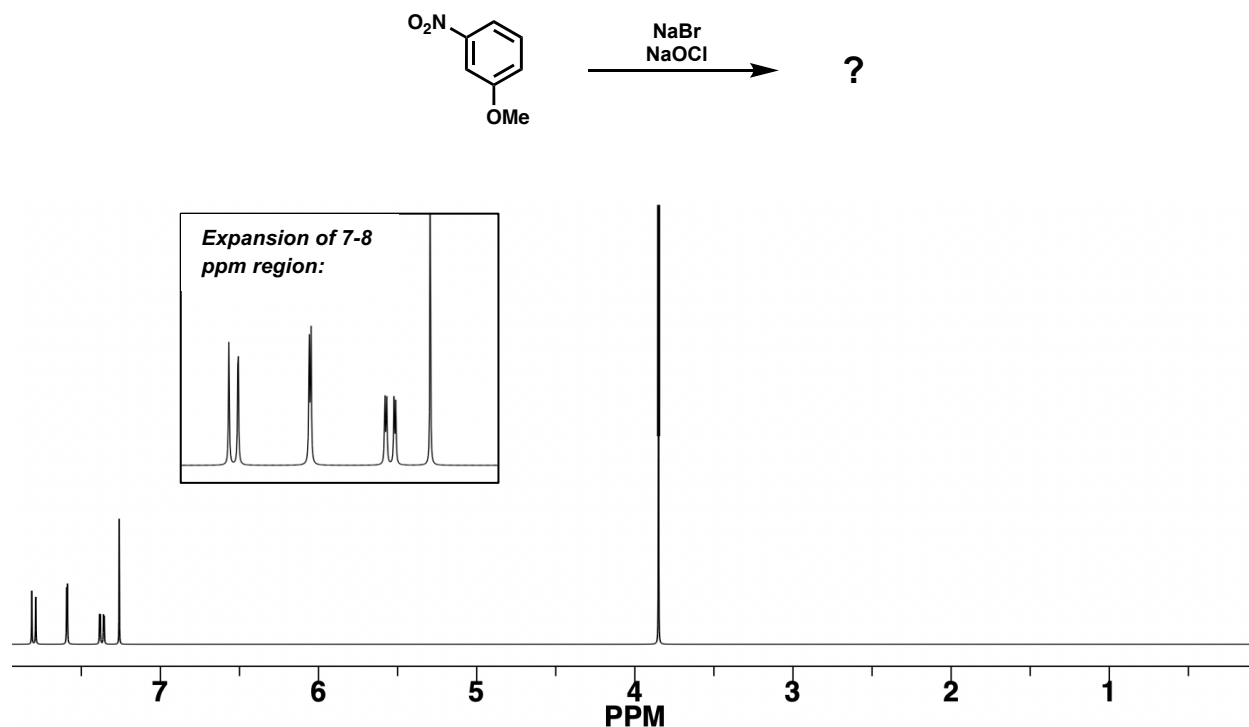
$\text{H}_\text{H} =$

$\text{H}_\text{D} =$

$\text{H}_\text{I} =$

$\text{H}_\text{E} =$

- 6) A ^1H NMR spectrum of the product of the monobromination reaction below is obtained in CDCl_3 :



A. Draw the product of this reaction.

B. Label each aromatic proton on your product structure. For each proton:

i) Predict the splitting pattern.

ii) Predict if the proton's signal will be upfield or downfield relative to benzene.

iii) Assign each proton to a signal in the NMR spectrum. Does your expected splitting pattern/chemical shift match the spectrum?